MEMORY ALLOCATION

1.FIRST FIT

CODE :

```
#include<stdio.h>
int main(){
        int n,m;
        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);
        int blockSize[m], remainingSize[m];
        printf("Enter size of each block:\n");
        for(int i=0;i<m;i++){
                scanf("%d",&blockSize[i]);
                remainingSize[i]=blockSize[i];
        }

        printf("enter number of processes:\n");
        scanf("%d",&n);
        int processSize[n];
        printf("Enter size of each process:\n");
        for(int i=0;i<n;i++){
                scanf("%d",&processSize[i]);
        }
        for(int i=0;i<n;i++){
                int allocated = 0;
                for(int j=0;j<m;j++){
                        if(remainingSize[j]>=processSize[i]){
                                int fragment = remainingSize[j] - processSize[i];
                                printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], j+1, blockSize[j],
 fragment);
                                remainingSize[j] -= processSize[i];
                                allocated = 1;
                                break;
                        }
                }
                if(!allocated){
                        printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
                }
        }
        return 0;
}
~
~
~
~
~
~
~
"firstfit.c" 37L, 959B                                                                                    37,0-1        All
```

OUTPUT :

```
[tejini@localhost ~]$ cd memory
[tejini@localhost memory]$ vi firstfit.c
[tejini@localhost memory]$ gcc firstfit.c
[tejini@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
[tejini@localhost memory]$
```

2.BEST FIT

CODE

Activities Terminal Apr 20 10:26

tejini@localhost:~/memory — /usr/bin/vim bestfit.c

```
#include<stdio.h>
int main(){
        int m,n;
        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);
        int blockSize[m], remaining[m];
        printf("Enter size of each block:\n");
        for(int i=0;i<m;i++)
        {
                scanf("%d",&blockSize[i]);
                remaining[i]=blockSize[i];
        }
        printf("Enter number of processes:\n");
        scanf("%d",&n);
        int processSize[n];
        printf("Enter size of each process:\n");
        for(int i=0;i<n;i++)
        {
                scanf("%d",&processSize[i]);
        }
        for(int i=0;i<n;i++){
                int bestIndex=-1;
                for(int j=0;j<m;j++){
                        if(remaining[j]>=processSize[i]){
                                if(bestIndex==-1 || remaining[j] < remaining[bestIndex]){
                                        bestIndex=j;
                                }
                        }
                }
                if(bestIndex != -1){
                        int fragment = remaining[bestIndex] - processSize[i];
                        printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n",i+1,processSize[i],bestIndex+1,blockSize[bestInd
ex],fragment);
                        remaining[bestIndex]-=processSize[i];
                }
                else{
                        printf("Process %d of size %d is not allocated\n",i+1,processSize[i]);
                }
        }
        return 0;
}
                                                                                                                    1,17          Top
```

OUTPUT :

```
[tejini@localhost memory]$ gcc bestfit.c
[tejini@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
[tejini@localhost memory]$
```